

Assignment 1, Harmonic Analysis

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Problem 1

(a)

$$\begin{aligned} M'_c(f)(x) &= \sup_{r>0} \frac{1}{r^d} \int_{x+(-r/2, r/2)^d} |f(y)| dy \\ &\leq \sup_{r>0} \frac{1}{r^d} \int_{B(x, r\sqrt{d}/2)} |f(y)| dy && (x + (-r/2, r/2)^d \subseteq B(x, r\sqrt{d}/2)) \\ &= \sup_{\rho>0} \frac{d^{d/2}}{2^d \rho^d} \int_{B(x, \rho)} |f(y)| dy && (\rho = r\sqrt{d}/2) \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \sup_{\rho>0} \frac{1}{v_d \rho^d} \int_{B(x, \rho)} |f(y)| dy \\ &= d^{d/2} \cdot v_d \cdot 2^{-d} \cdot M_c(f)(x). \end{aligned}$$

(b)

$$\begin{aligned} M_c(f)(x) &= \sup_{r>0} \frac{1}{v_d r^d} \int_{B(x, r)} |f(y)| dy \\ &\leq \sup_{r>0} \frac{1}{v_d r^d} \int_{x+(-r, r)^d} |f(y)| dy && (B(x, r) \subseteq x + (-r, r)^d) \\ &= \sup_{s>0} \frac{2^d}{v_d s^d} \int_{x+(-s/2, s/2)^d} |f(y)| dy && (s = 2r) \\ &= 2^d \cdot v_d^{-1} \sup_{s>0} \frac{1}{s^d} \int_{x+(-s/2, s/2)^d} |f(y)| dy \\ &= 2^d \cdot v_d^{-1} \cdot M'_c(f)(x). \end{aligned}$$

Problem 2

(a)

Since clearly $g_q(x) := |f(x) - q| \in L^1_{\text{loc}}(\mathbb{R}^d)$, the Lebesgue Differentiation Theorem gives

$$\lim_{r \rightarrow 0^+} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - q| dy = |f(x) - q| \quad \text{for a.e. } x \in \mathbb{R}^d.$$

Let A_q be the null set where this fails. Then $|A_q| = 0$ and the limit holds for all $x \in \mathbb{R}^d \setminus A_q$.

(b)

(c)

Problem 3